



Held for a Tap, Sent Regardless

A Street-Deployed Triage Bot for Doorbell Person-at-the-Gate Alerts, and What the Confirm Window Actually Changes

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Abstract. A doorbell camera’s “person at the gate” alert reaches a street’s neighbourhood-watch group chat within seconds of the sensor firing, whether or not anyone at the gate was worth flagging. We design and deploy a confirm-gate: a lightweight relay that holds each alert in a private volunteer queue for a fixed window, during which any rostered volunteer may tap to confirm before the alert posts to the group; alerts nobody confirms in time post anyway, carrying an “unconfirmed” tag instead of a “confirmed” one. Across eleven streets over a nine-week deployment, only 18.3% of alerts are confirmed inside the window, and weekly forwarding volume to the group chat is statistically indistinguishable from the pre-deployment baseline. Confirm latency runs longer, not shorter, on streets with larger volunteer rosters. An ablation removing the roster’s rotation-reminder ping changes the confirm-within-window rate by less than a percentage point. A follow-up member survey finds recall of an alert’s own tag performs no better than chance. We interpret the confirm-gate as adding a credibility marker to the alert stream without measurably changing what the stream delivers, and discuss the result alongside existing findings on alarm fatigue, bystander diffusion of responsibility, and warning-label effects.

Introduction

A commercial doorbell camera’s motion-triggered “person at the gate” alert is designed to travel fast: sensor to owner’s phone in under a second, owner’s phone to the street’s neighbourhood-watch group chat in whatever time it takes someone to forward a screenshot. Fast is also the problem. A meaningful share of these alerts concern a courier, a returning resident, or a cat, and the group chat that receives all of them regardless has no institutional memory of which alerts turned out to matter and which didn’t [1]. Prior work on alarm systems in clinical and meteorological settings documents what happens once a stream like this accumulates enough false positives: recipients recalibrate trust downward, a pattern this literature calls the cry-wolf effect [2], [3], [4]. A street group chat has no equivalent of a hospital’s alarm-management committee, but it does have volunteers.

We ask whether inserting one of those volunteers between the sensor and the chat — specifically, a rostered confirmation step with a short window to act — changes what actually reaches the street, or only changes what the street is told about what reached it. We design and deploy a confirm-gate: a small relay service that intercepts a doorbell’s person-detection webhook, notifies a rostered volunteer queue, and posts to the group chat only once a volunteer confirms or the window elapses, tagging the post accordingly either way.

Our contributions are threefold, each modest by design. First, we report a working, street-deployable confirm-gate architecture that requires no change to the underlying doorbell hardware or the group chat platform, which we expect generalises to other sensor-to-chat alert pipelines with only minor adaptation. Second, we measure confirm-within-window rate, confirm latency, and forwarding volume across

a nine-week, eleven-street deployment, and find the gate’s effect on what is delivered smaller than its effect on how delivery is labelled. Third, we relate roster size to confirm latency and find a direction — larger rosters confirm more slowly, not faster — that echoes findings on diffusion of responsibility in other online helping contexts, suggesting the confirm-gate inherits a known failure mode of volunteer coordination rather than avoiding it.

Related work

Smart home cameras generate alert volumes their owners routinely cannot sustain attending to; recent design work catalogues the resulting flood and proposes filtering strategies at the device level [1]. Doorbell-linked community platforms extend that flood past the household: Calacci et al. document how a major doorbell manufacturer’s companion app turned household alerts into a de facto, unaccountable neighbourhood surveillance network shared with police departments [5], and Kurwa traces a comparable pattern on a text-first neighbourhood platform, where an unmoderated alert stream became a vector for residents’ own suspicion rather than a record of verified incidents [6]. Neither line of work evaluates a confirmation step of the kind we deploy.

The alarm-fatigue literature is longer-established outside the home: clinical monitor alarms and severe-weather warnings both show that recipients recalibrate trust downward as false-positive rate rises, sometimes to the point of ignoring a genuine alarm outright [2], [3], [4], [7]. Adding a confirmation step is one candidate remedy; a structurally similar remedy, capping how far a message can be forwarded on a large messaging platform, delayed but did not stop the spread of the misinformation it targeted [8], and false stories have been found to spread further and faster than true ones on a comparable platform regardless of any friction placed in their path [9]. Separately, warning-label research finds that tagging a subset of claims can shift how untagged claims are read, an “implied truth effect” in which the presence of any label, not its content, carries the signal [10]. Whether a volunteer confirms at all is its own open question: online helping research finds that a would-be helper’s felt responsibility falls as the pool of other possible helpers grows, both in email requests [11], [12] and, with prim-

ing as a moderator, in more recent replications [13]. Wikipedia’s own volunteer-patrol literature shows a comparable coordination problem can nonetheless resolve fast when a vandal is visible and unambiguous [14], though it says little about what happens when the thing being patrolled is, most of the time, a parcel courier. Prior work at this institution found a comparable pattern in its own early-warning apparatus: the Horizon Register’s automated alert threshold tracked reporting behaviour more reliably than the priority it was built to detect [15], and a companion study of a neighbourhood giveaway group found only slight agreement between an automated flag and a volunteer moderator’s independent read [16].

Method

The confirm-gate sits between a doorbell camera’s motion-detection webhook and the street’s existing group chat, requiring accounts on neither: it holds a webhook token for the camera and a bot account for the chat. On a person-detected event, the gate (1) timestamps the alert, (2) posts a one-tap confirm link to a private queue visible only to that street’s rostered volunteers, and (3) starts a fixed window.

```
on person_detected(alert):
    t0 = now()
    notify(roster[street], alert, t0)
    loop until now() - t0 >= WINDOW:
        if alert.confirmed: break
    if alert.confirmed:
        post(chat[street], alert,
tag="confirmed", by=alert.confirmer)
    else:
        post(chat[street], alert,
tag="unconfirmed")
```

No alert is ever suppressed outright — the design brief from participating streets was explicit that a missed genuine event was a worse outcome than a mistagged one, so the gate’s only lever is the tag, never the delivery.

We model the probability that at least one of a roster of n volunteers confirms within a window of length w as a competing-exponentials process, each volunteer treated as an independent Poisson responder with a per-volunteer response hazard λ estimated from each street’s own app-open telemetry over the baseline weeks:

$$P(\text{confirm by } w) = 1 - e^{-n\lambda w}.$$

The model predicts confirm-within-window rate rising with roster size; the Results below test this prediction against the deployed rosters.

Eleven streets already running a volunteer-rostered neighbourhood-watch group chat took part, recruited through the School's existing community-liaison contacts and screened only for having an active roster of two or more volunteers. Each street ran three baseline weeks (alerts forwarded unmodified, as before) followed by six weeks with the confirm-gate live, a 90-second window chosen from a pilot week's median volunteer app-open latency. We logged every alert's timestamp, tag, and, when applicable, confirm latency, and administered a two-question end-of-deployment survey to each street's group-chat members asking them to recall the tag on the most recent alert they remembered seeing.

Results

Across the six-week confirm-gate phase, the eleven streets logged 930 person-detected alerts, of which 170 (18.3%) were confirmed inside the 90-second window. Figure 1 tracks mean alerts forwarded per street across the full nine-week deployment; the confirm-gate goes live at the week 3/4 boundary and the series shows no visible step change (baseline $M = 14.6$ alerts/week/street, confirm-gate $M = 14.1$, paired $t(10) = 0.38$, $p = .71$, $d = 0.06$).

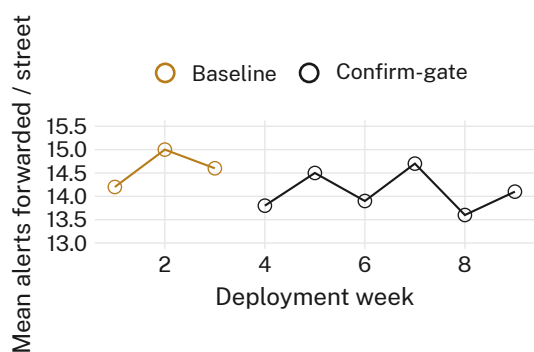


Figure 1: Mean alerts forwarded per street held steady across the baseline/confirm-gate boundary (14.6 vs. 14.1/week, n.s.) — the gate changes how an alert is tagged, not whether it arrives.

We had modelled confirm-within-window rate as increasing with roster size (Method, above); the deployed rosters show the opposite. Figure 2 plots confirm latency by roster-size cohort: streets with rosters of three or fewer volunteers

confirmed at a median 55 seconds (IQR 36–72, within the window for the clear majority of attempts), while streets with rosters of six or more confirmed at a median 136 seconds (IQR 94–177, past the window for the clear majority). We do not observe roster sizes in our sample large enough to test the model's predicted asymptote, but the direction it predicted does not appear in the range we could deploy.

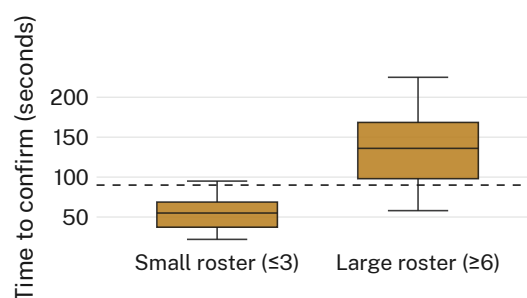


Figure 2: Confirm latency ran longer, not shorter, on streets with larger volunteer rosters (median 55s vs. 136s), against a 90-second window (dashed).

An ablation removed the rotation-reminder ping — a notification nudging whichever volunteer was next in the roster's rotation to check the queue — from four streets for the deployment's final two weeks. Confirm-within-window rate on those streets was 17.9%, against 18.7% on the seven streets that kept the ping ($\chi^2(1) = 0.09$, $p = .77$); we retain the ping for its unmeasured goodwill value.

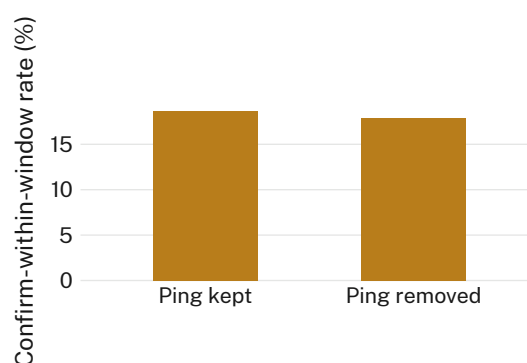


Figure 3: Removing the roster's rotation-reminder ping left the confirm-within-window rate unchanged (18.7% vs. 17.9%, n.s.).

The end-of-deployment survey returned 96 responses across the eleven group chats. Asked to recall whether the most recent alert they remembered had been tagged "confirmed" or "unconfirmed," 21% answered correctly; among

the 68% who offered a guess rather than declining to answer, accuracy (49%) did not exceed the rate expected by chance.

Discussion

The confirm-gate changes what an alert is called far more reliably than it changes whether an alert arrives. Forwarding volume held steady across the deployment; the tag, not the alert, is the gate's real output. This sits alongside the implied truth effect documented in warning-label research, where attaching a marker to a subset of claims shifts how the unmarked remainder is read [10] — our survey result, chance-level recall of which tag an alert actually carried, suggests the street audience is not reading the marker closely enough for even that shift to register, which may be the more comfortable outcome for a system that never suppresses.

The roster-size finding is consistent with diffusion-of-responsibility accounts of online helping behaviour, where a larger pool of potential responders lowers, rather than raises, any one member's felt responsibility to act [11], [12], [13]. A larger volunteer roster reads as more capacity on paper and less capacity in the confirm queue, especially once checking it is treated as the kind of task-switching interruption whose resumption cost the broader interruption literature has already priced in [17]. We observe consistent slowing across every large-roster street in the sample, which we read as suggestive rather than definitive given the number of streets involved.

Limitations

Our deployment covers eleven streets and nine weeks, a modest base for the roster-size comparison in particular; the consistency of the slowing direction across every large-roster street, however, is not the pattern a sampling artefact would be expected to produce. The 90-second window was fixed from a single pilot week's telemetry rather than tuned per street, which likely understates what a locally calibrated window could achieve. We measure recall of an alert's tag, not downstream behaviour change, so we cannot rule out that the tag shapes action even where it fails to register consciously — a stronger claim would need a behavioural outcome the present design does not capture.

Conclusion

We built and deployed a confirm-gate that inserts a rostered volunteer's tap between a doorbell camera's person-detection alert and a street's group chat, and found that across eleven streets and nine weeks it relabels alerts far more reliably than it filters them: forwarding volume held steady, confirm-within-window rate stayed under one in five, and larger volunteer rosters confirmed slower rather than faster. The tag itself went largely unread. Future work might calibrate the window per street, test a genuinely suppressive gate against the participating streets' stated preference against one, or extend the roster-size comparison past the range this deployment could reach.

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